Sharp Concentration of Positive Best m-Term Error under Cone Measure

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Status and source question

Status: candidate full answer, likely valid, pending expert review. Let μ_p be normalized cone measure on the positive part

$$\Delta_p^n = \{x \in \mathbb{R}_+^n : \textstyle\sum_{i=1}^n x_i^p = 1\} \qquad (0 < p < \infty),$$

and write $x_1^* \geq \cdots \geq x_n^*$ for the decreasing rearrangement. Vybíral [1] proves the mean estimate

$$\sigma_m^{p,\infty}(\mu_p) = \mathbb{E}_{\mu_p} x_{m+1}^* \asymp_p \left[\frac{\log(en/(m+1))}{n}\right]^{1/p}$$

for m at most a fixed fraction of n. Remark 1(i), source PDF page 16, asks for a concentration statement analogous to the known $m = 0$ theorem of Schechtman–Zinn [2], for every $m > 0$.

gives the result. $\square$

Remark 1. (i) Theorem 7 provides basic estimates of average best m-term widths $\sigma_m^{p,\infty}(\mu_p)$. In the case $m = 0$ a stronger result on concentration of μ_p was obtained already in [43, Theorem 3 and Remark 2]. It would be certainly of interest to obtain a similar statement also for other values of $m > 0$, but this would go beyond the scope of this paper and we leave this direction open for further study.

Putting $k = m + 1$, the theorem below gives matching two-sided tail exponents throughout the entire nontrivial upper-tail range and an explicit high-probability window at the mean scale.

Proof intuition

Cone measure turns the dependent coordinates into normalized independent Gamma variables: $X_i^p = Y_i/(Y_1 + \cdots + Y_n)$ with $Y_i \sim \mathrm{Gamma}(1/p, 1)$. The denominator is exponentially concentrated around a constant multiple of n. Above that radial event, $X_k^* > tn^{-1/p}$ means that at least k independent Gamma variables exceed a constant multiple of t^p. The number of possible k-subsets contributes $k\log(en/k)$, explaining both the threshold $t^p \asymp \log(en/k)$ and the upper exponent $-\Theta_p(kt^p)$.

For the reverse inequality, choose one fixed k-subset, force those Gamma variables into a short interval of height $\Theta_p(t^p)$, and keep the remaining radial sum below $O_p(n)$. The deterministic condition $kt^p \lesssim_p n$ ensures normalization cannot push the selected coordinates below threshold. This event already has probability $\exp[-O_p(kt^p)]$, matching the upper exponent.

Main result

Theorem 1 (Matching concentration tails)**.** *Fix $0 < p < \infty$. There exist constants $A_p > 1$, $b_p, c_p, C_p > 0$ and $\varepsilon_p \in (0, 1/2)$ such that the following holds. Let $n \geq 2$, $1 \leq k \leq \varepsilon_p n$, and put*

$$L_{n,k} = \log \frac{en}{k}.$$

For every t satisfying

$$A_p L_{n,k}^{1/p} \leq t \leq b_p (n/k)^{1/p}, \tag{1}$$

one has

$$\exp(-C_p k t^p) \leq \mu_p\{x \in \Delta_p^n : n^{1/p} x_k^* > t\} \leq \exp(-c_p k t^p). \tag{2}$$

Thus the Schechtman–Zinn $k = 1$ exponent extends to every positive best-term index, with the sharp multiplicative factor $k = m + 1$.

Corollary 2 (A high-probability window)**.** *After changing the constants, for every $1 \leq k \leq \varepsilon_p n$,*

$$\mu_p\left\{ x_k^* > A_p \left(\frac{L_{n,k}}{n}\right)^{1/p} \right\} \leq C_p e^{-c_p k L_{n,k}},$$
$$\mu_p\left\{ x_k^* < a_p \left(\frac{L_{n,k}}{n}\right)^{1/p} \right\} \leq C_p e^{-c_p \sqrt{nk}}$$

for a constant $a_p > 0$. In particular x_{m+1}^* is concentrated, up to fixed p-dependent factors, at the mean scale from source Theorem 7.

Theorem 3 (Exact $p = \infty$ law)**.** *Let μ_∞ be cone measure on the positive ℓ_∞^n sphere. For $1 \leq k \leq n$ and $0 \leq r < 1$,*

$$\mu_\infty\{x_k^* > r\} = \mathbb{P}\{\mathrm{Bin}(n-1, 1-r) \geq k-1\}.$$

Hence all concentration estimates in this remaining endpoint are exactly the standard binomial Chernoff bounds.

Proof

We first record the only one-dimensional estimates needed.

Lemma 4 (Gamma tails)**.** *Let $Y \sim \mathrm{Gamma}(\alpha, 1)$ with fixed $\alpha > 0$. There are positive constants depending only on α such that, for $u \geq 1$,*

$$\mathbb{P}(Y > u) \leq C_\alpha e^{-c_\alpha u}, \qquad \mathbb{P}(Bu \leq Y \leq 2Bu) \geq e^{-C_{\alpha,B} u}$$

for each fixed $B \geq 1$. Also $\mathbb{P}(Y > u) \geq c_\alpha e^{-C_\alpha u}$ for all $u \geq 0$. If S_N is a sum of N such variables, then for suitable $0 < d_\alpha < D_\alpha$,

$$\mathbb{P}(S_N < d_\alpha N) + \mathbb{P}(S_N > D_\alpha N) \leq 2e^{-c_\alpha N}.$$

Proof. The density is $y^{\alpha-1}e^{-y}/\Gamma(\alpha)$. Integrating it on $[u, \infty)$ and on a unit subinterval of $[Bu, 2Bu]$ gives the first three bounds, since every fixed power of u is between two exponentials at this scale. The last display is the standard Chernoff bound for $S_N \sim \mathrm{Gamma}(N\alpha, 1)$. □

Proof of Theorem 1. The cone-measure representation used in the source gives independent $Y_i \sim \mathrm{Gamma}(1/p, 1)$ such that

$$X_i^p = \frac{Y_i}{S}, \qquad S = \sum_{j=1}^n Y_j, \tag{3}$$

and $X = (X_1, \ldots, X_n)$ has law μ_p.

For the upper bound, work first on $S \geq d_p n$. If $n^{1/p} X_k^* > t$, then at least k of the Y_i exceed $d_p t^p$. Lemma 4 and a union bound give

$$\begin{aligned} \mathbb{P}\{Y_k^* > d_p t^p\} &\leq \binom{n}{k} \mathbb{P}\{Y_1 > d_p t^p\}^k \\ &\leq \exp\{k L_{n,k} - c_p k t^p\}. \end{aligned}$$

If A_p in (1) is sufficiently large, this is at most $\exp(-c_p' k t^p)$. The discarded probability $\mathbb{P}(S < d_p n)$ is $e^{-c_p n}$. Since the upper restriction in (1) gives $kt^p \leq b_p^p n$, this radial error is absorbed into the same exponential after decreasing b_p or c_p'.

For the lower bound, choose D_p so that

$$\mathbb{P}\left\{ \sum_{i=k+1}^n Y_i \leq D_p n \right\} \geq \frac{1}{2}$$

(Markov's inequality already suffices), and set $B_p = 4D_p$. Consider

$$E = \{B_p t^p \leq Y_i \leq 2B_p t^p \ (1 \leq i \leq k)\} \cap \left\{ \sum_{i=k+1}^n Y_i \leq D_p n \right\}.$$

Choose $b_p^p \leq 1/8$. On E and in the range (1),

$$S \leq 2B_p k t^p + D_p n \leq \frac{B_p}{2} n, \qquad \frac{n Y_i}{S} \geq 2t^p \quad (1 \leq i \leq k).$$

Thus $E \subset \{n^{1/p} X_k^* > t\}$. Independence and Lemma 4 give

$$\mathbb{P}(E) \geq \frac{1}{2} \mathbb{P}\{B_p t^p \leq Y_1 \leq 2B_p t^p\}^k \geq e^{-C_p k t^p},$$

after enlarging C_p. This proves (2). □

Proof of Corollary 2. For the upper estimate, repeat the union-bound argument with $t = A_p L_{n,k}^{1/p}$ and choose A_p so large that $\mathbb{P}(Y_1 > d_p A_p^p L_{n,k}) \leq e^{-3L_{n,k}}$. Then the binomial union bound is at most $e^{-2kL_{n,k}}$; the lower radial tail is absorbed because $kL_{n,k} \leq n$.

For the lower estimate, work on $S \leq D_p n$. Put $u = D_p a_p^p L_{n,k}$ and let $N_u = \#\{i : Y_i > u\}$. If $N_u \geq k$, then $X_k^* > a_p (L_{n,k}/n)^{1/p}$. By Lemma 4, a_p can be chosen so small that

$$q := \mathbb{P}(Y_1 > u) \geq c_p e^{-L_{n,k}/2} = c_p e^{-1/2} \sqrt{k/n}.$$

After choosing ε_p small, $nq \geq 4k$. A binomial Chernoff bound therefore yields

$$\mathbb{P}(N_u < k) \leq e^{-c_p n q} \leq e^{-c_p' \sqrt{nk}}.$$

The upper radial tail is $e^{-c_p n}$ and is absorbed into this bound. □

Proof of Theorem 3. Let $U_1, \dots, U_n$ be independent uniform variables on $[0,1]$ and put $M = \max_i U_i$. Radial projection U/M has cone measure μ_∞. Conditionally on the almost surely unique maximizing index and on M, the remaining $n-1$ ratios U_i/M are independent uniform variables on $[0,1]$. The maximum ratio is 1, so the kth largest ratio exceeds r exactly when at least $k-1$ of those $n-1$ uniforms exceed r. □

Verification, scope, and novelty

The proof is symbolic; the included Monte Carlo script only checks the scale across $p \in \{1/2, 1, 2\}$, dimensions, and positive m. The result answers the source sentence tied to $\sigma_m^{p,\infty}(\mu_p)$ and Theorem 7. It does not claim concentration for Hausdorff surface measure or for finite target exponent q.

The run indexes and bounded exact-title, arXiv-id, best-m-term concentration, cone-measure order-statistic, and kth-coordinate searches through 11 August 2026 found no matching later resolution. General independent-order-statistic concentration results exist, but the coordinates in (3) share the random denominator; no searched source recorded the normalized theorem or the matching exponent above. Novelty confidence is moderate, subject to specialist citation review.

Human-review recommendation

Check the constants in the deterministic implication on the lower event E, especially $B_p = 4D_p$ and $b_p^p \le 1/8$. Then check that both discarded Gamma-sum tails are absorbable in the stated ranges. Those are the only places where the independent order statistic and the normalized cone vector differ materially.

References

[1] J. Vybíral, *Average best m-term approximation*, Constr. Approx. 36 (2012), 83–115; arXiv:1009.1751.

[2] G. Schechtman and J. Zinn, *On the volume of the intersection of two L_p^n balls*, Proc. Amer. Math. Soc. 110 (1990), 217–224; arXiv:math/9201206.